K.R. MANGALAM UNIVERSITY

Bachelor of Technology in Computer Science and Engineering (AI & ML)

Lab File

Computer Networks Lab

Course Code: ENCS352

SUBMITTED TO:

Ms. Jyoti Kaurav

Assistant Professor

SOET

SUBMITTED BY:

Akshita Goel

2301730231

Section: D

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PRACTICAL 1(A)

CPT Platform Overview

Objective:

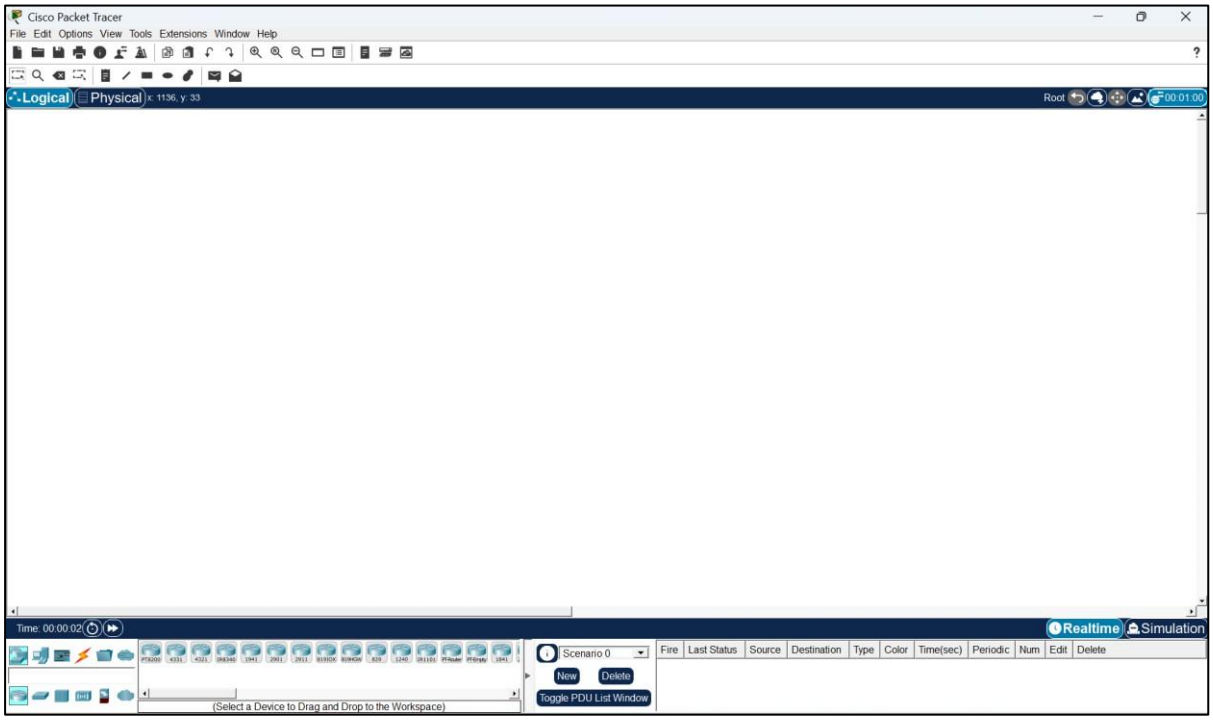
To understand the interface and components of Cisco Packet Tracer and become familiar with its working environment.

Apparatus:

Computer with Cisco Packet Tracer 9.0 installed

Theory:

Cisco Packet Tracer is a network simulation tool by Cisco Systems. It allows designing and testing network topologies without physical hardware. The interface has a Menu Bar at top with File, Edit, Options, Tools. Below is the Toolbar for Save, Open and Zoom. The large white area is the Workspace where devices are placed. At the bottom left is the Device Panel showing Routers, Switches, End Devices. The Cable Panel provides Straight-Through, Crossover and Serial cables. Bottom right has Realtime and Simulation mode toggle. The PDU List shows packet details during simulation.



The Packet Tracer interface has these main parts:

Area	**Description**
Menu Bar	Contains options like File, Edit, Options, View, Tools
Toolbar	Provides quick access to common functions like Save, Open
Workspace	Main area to design network topology
Device Panel	Contains networking devices like routers, switches
Cable Panel	Provides different cable types for connections
Realtime/Simulation	Used to observe packet flow step-by-step
PDU List	Displays packet details during simulation

Result:

The Cisco Packet Tracer interface was successfully explored. All panels and components were identified as shown in the screenshot.

Conclusion:

Cisco Packet Tracer provides a complete virtual environment for network simulation without any physical hardware requirement.

PRACTICAL 1(B)

Peer to Peer Network

Objective:

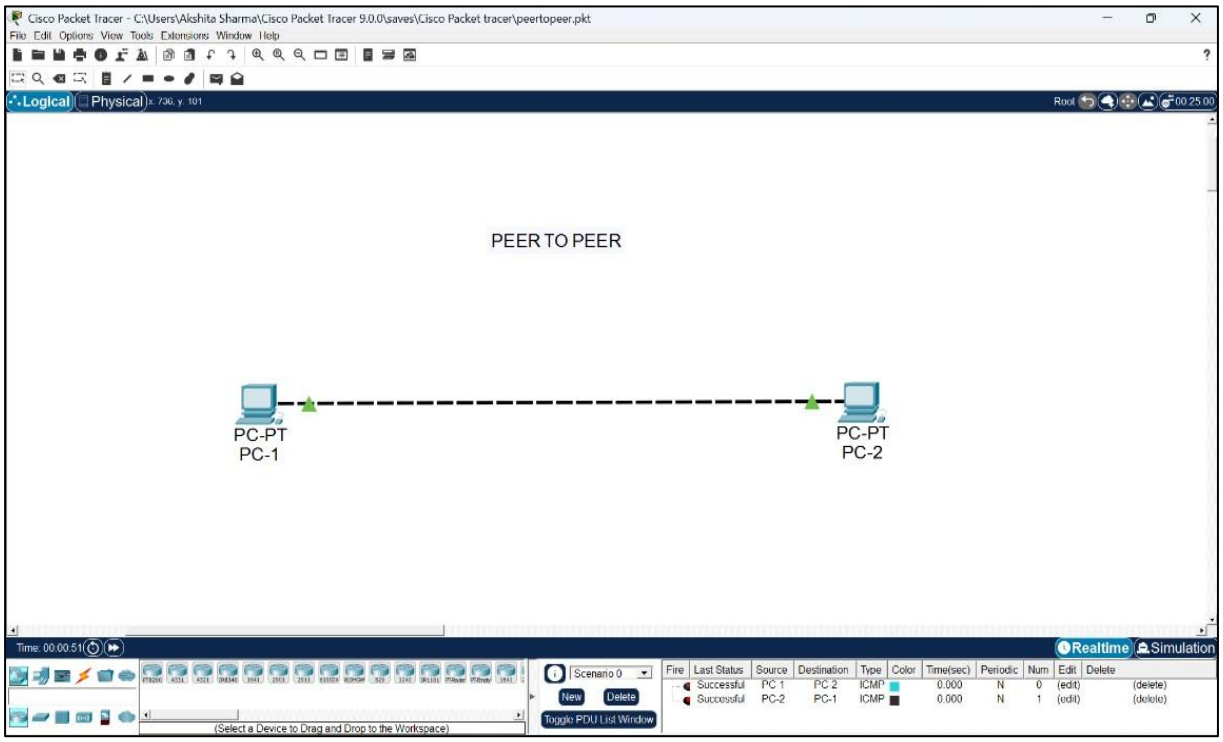
To create a peer-to-peer network by directly connecting two PCs and verify communication between them.

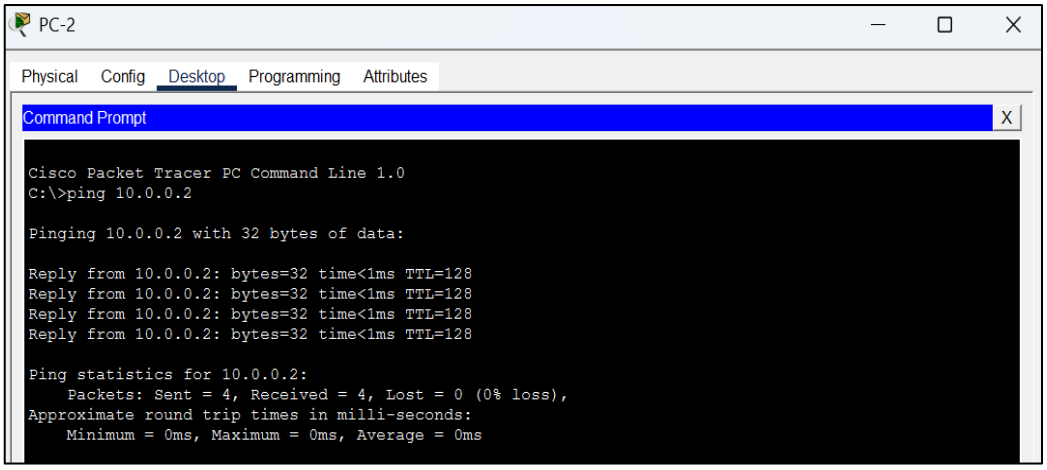
Theory:

A Peer-to-Peer (P2P) network is the most basic type of network where two or more computers are connected directly to each other without any central server, switch or hub. Each computer acts as both a client and a server simultaneously. A crossover cable is used to directly connect two PCs because both devices are of the same type. In P2P networks, files and resources can be shared directly between computers.

Apparatus / Software Required:

Cisco Packet Tracer, 2 PCs, 1 Copper Crossover Cable





Result:

As seen in the screenshot, ping from PC-2 to 10.0.0.2 was successful. 4 packets sent, 4 received, 0% packet loss. Simulation mode confirmed successful ICMP packets between PC-1 and PC-2.

Conclusion:

Peer-to-peer network was successfully created. Direct communication between two PCs was achieved using a crossover cable.

PRACTICAL 2(A)

NETWORK TOPOLOGIES

Star Topology

Objective:

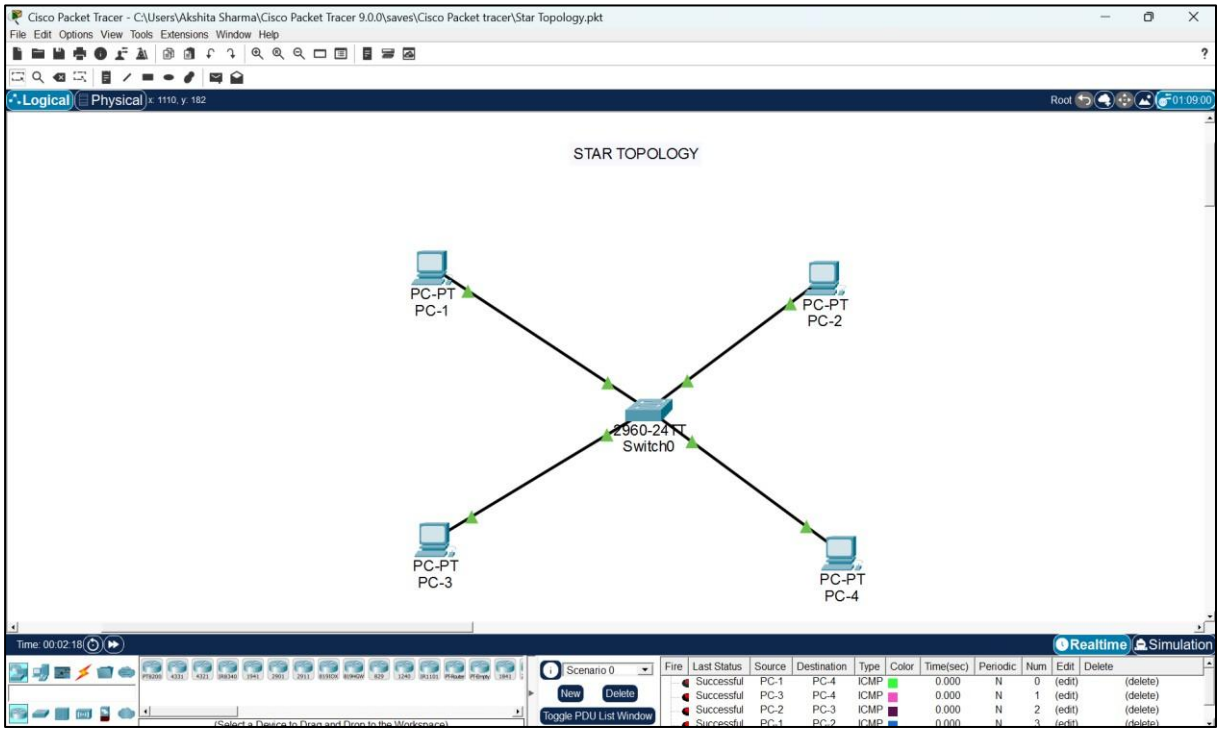
To design and implement a Star topology network in Cisco Packet Tracer and verify connectivity between all devices.

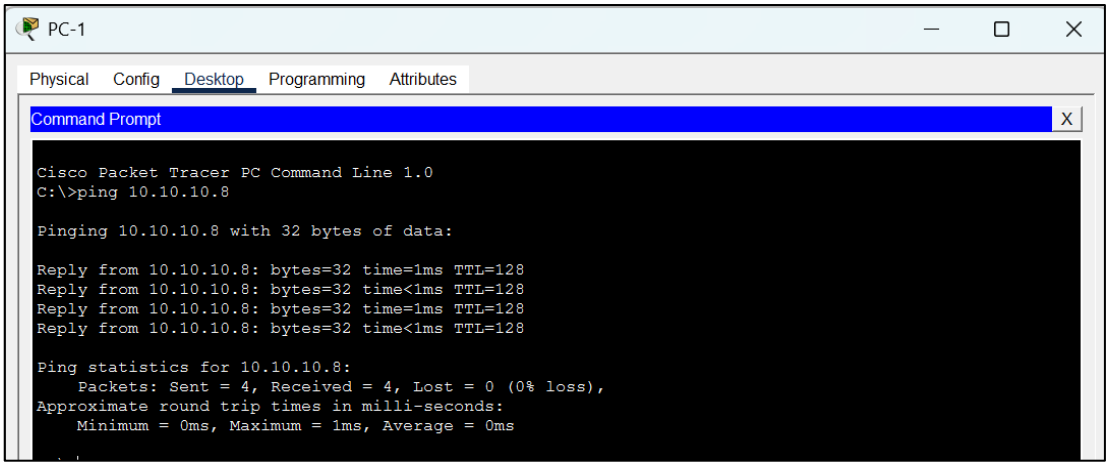
Theory:

In Star topology all devices are connected to a central device which is usually a switch or hub. All communication between devices passes through this central device. If one device fails it does not affect other devices. However if the central switch fails the entire network goes down. Star topology is the most widely used topology in modern LANs.

Apparatus / Software Required:

Cisco Packet Tracer, 4 PCs, 1 Switch 2960, Straight-through cables





Result:

As seen in the screenshot, star topology was successfully built with Switch0 at center and PC-1, PC-2, PC-3, PC-4 connected. Ping from PC-1 to 10.10.10.8 was successful. Simulation showed successful ICMP packets between all PCs.

Conclusion:

Star topology was successfully implemented. It is the most reliable topology for LANs as failure of one device does not affect others.

PRACTICAL 2(B)

NETWORK TOPOLOGIES

Ring Topology

Objective:

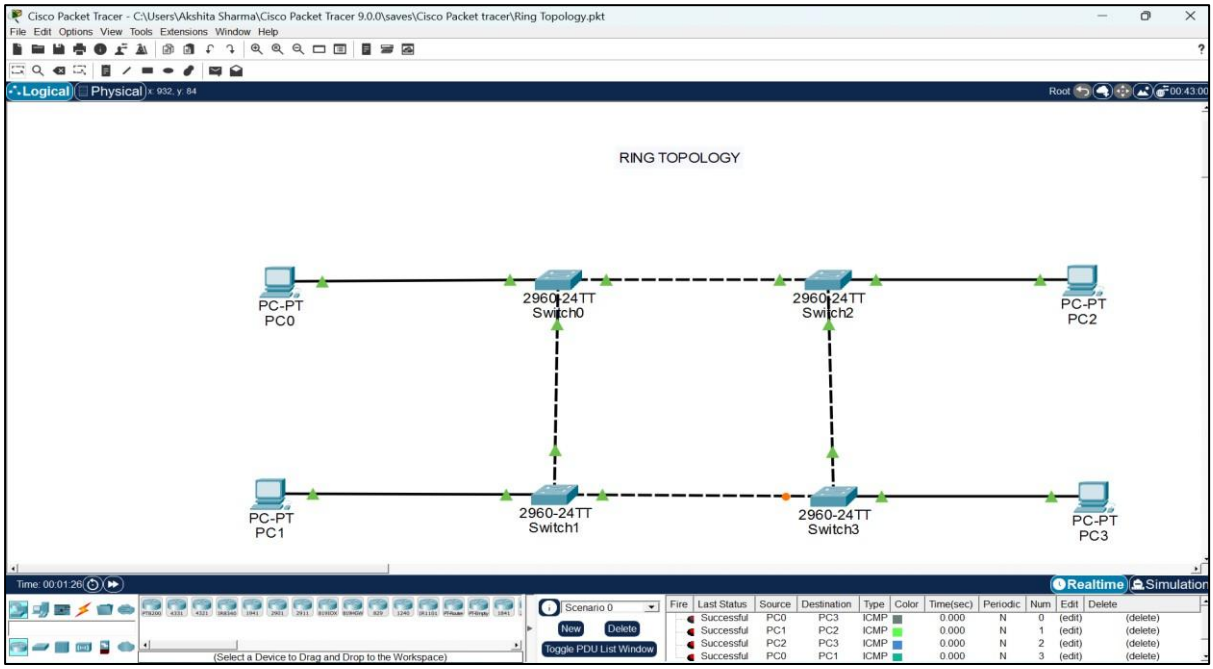
To design and simulate a Ring topology using switches and PCs in Cisco Packet Tracer.

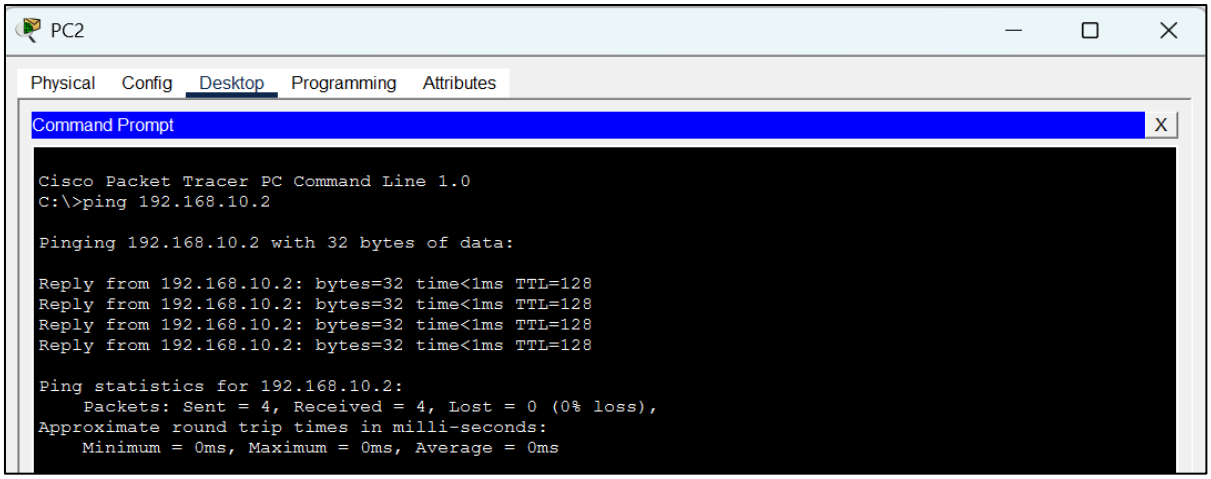
Theory:

In Ring topology each device is connected to exactly two other devices forming a closed circular loop. Data travels in one direction around the ring. Each device acts as a repeater. If one device or cable fails the entire network can be disrupted. Ring topology is used in some WANs and token ring networks.

Apparatus / Software Required:

Cisco Packet Tracer, 4 PC-PT (PC0, PC1, PC2, PC3), 4 Switches 2960-24TT (Switch0, Switch1, Switch2, Switch3), Crossover cables





Result:

Ring topology was successfully built as seen in the screenshot. 4 switches connected in a ring with one PC each. Ping from PC2 to 192.168.10.2 was successful. Simulation showed all ICMP packets as Successful.

Conclusion:

Ring topology was successfully implemented using 4 switches connected in a circular arrangement with PCs attached to each switch.

PRACTICAL 2(C)

NETWORK TOPOLOGIES

Mesh Topology

Objective:

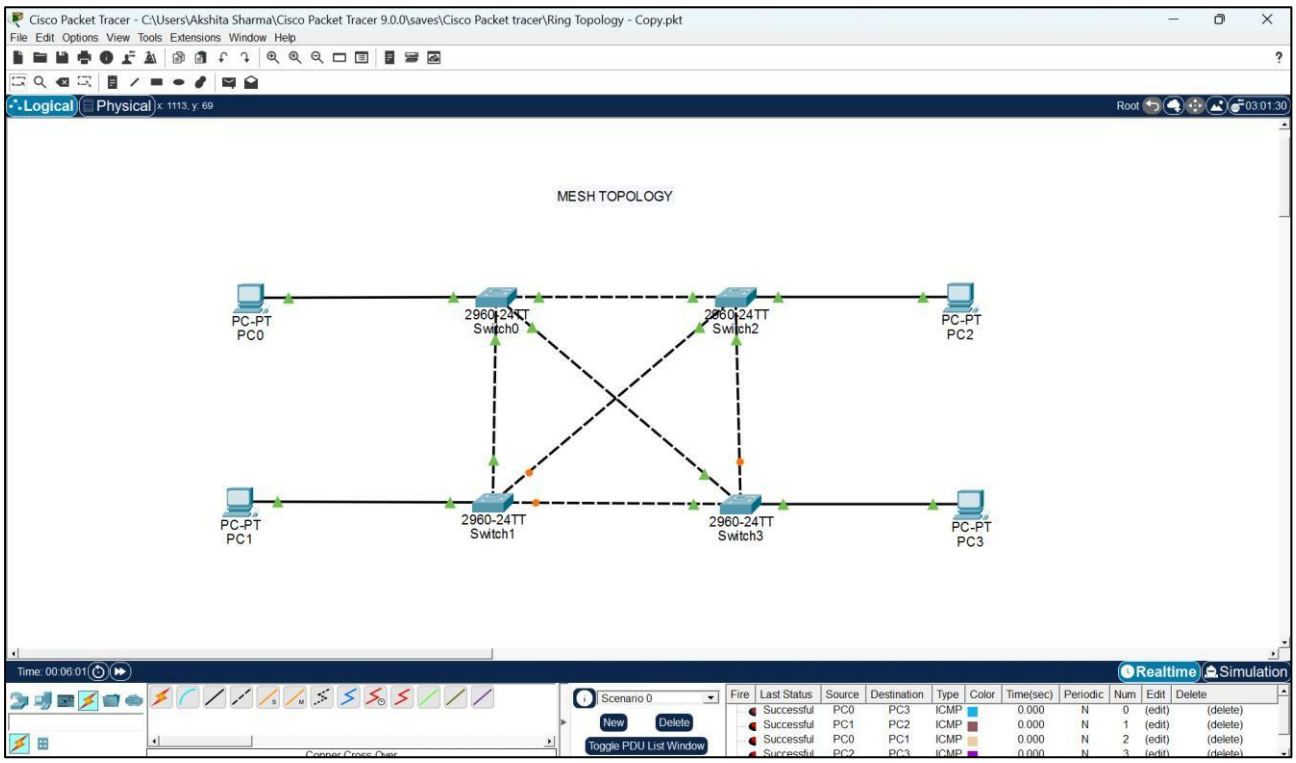
To design and implement a Mesh topology network in Cisco Packet Tracer.

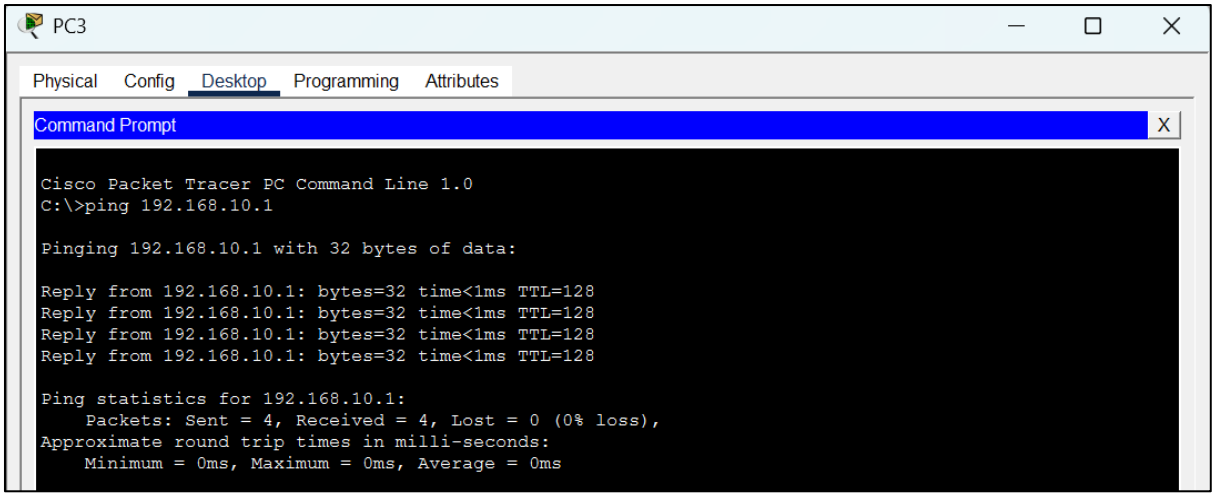
Theory:

In Mesh topology every device is connected to every other device in the network. It provides maximum redundancy and reliability. If one path fails data can be sent through another path. There are two types — Full Mesh where every device connects to every other device and Partial Mesh where only some devices have multiple connections. Mesh is used in WANs, military networks and critical infrastructure.

Apparatus / Software Required:

Cisco Packet Tracer, 4 PC-PT (PC0, PC1, PC2, PC3), 4 Switches 2960-24TT, Crossover cables





Result:

Mesh topology was successfully built as seen in screenshot. All switches are interconnected with crossover cables providing multiple paths. Ping from PC3 to 192.168.10.1 was successful with 0% packet loss.

Conclusion:

Mesh topology was successfully implemented. It provides maximum redundancy as there are multiple paths between any two devices.

PRACTICAL 2(D)

NETWORK TOPOLOGIES

Tree Topology

Objective:

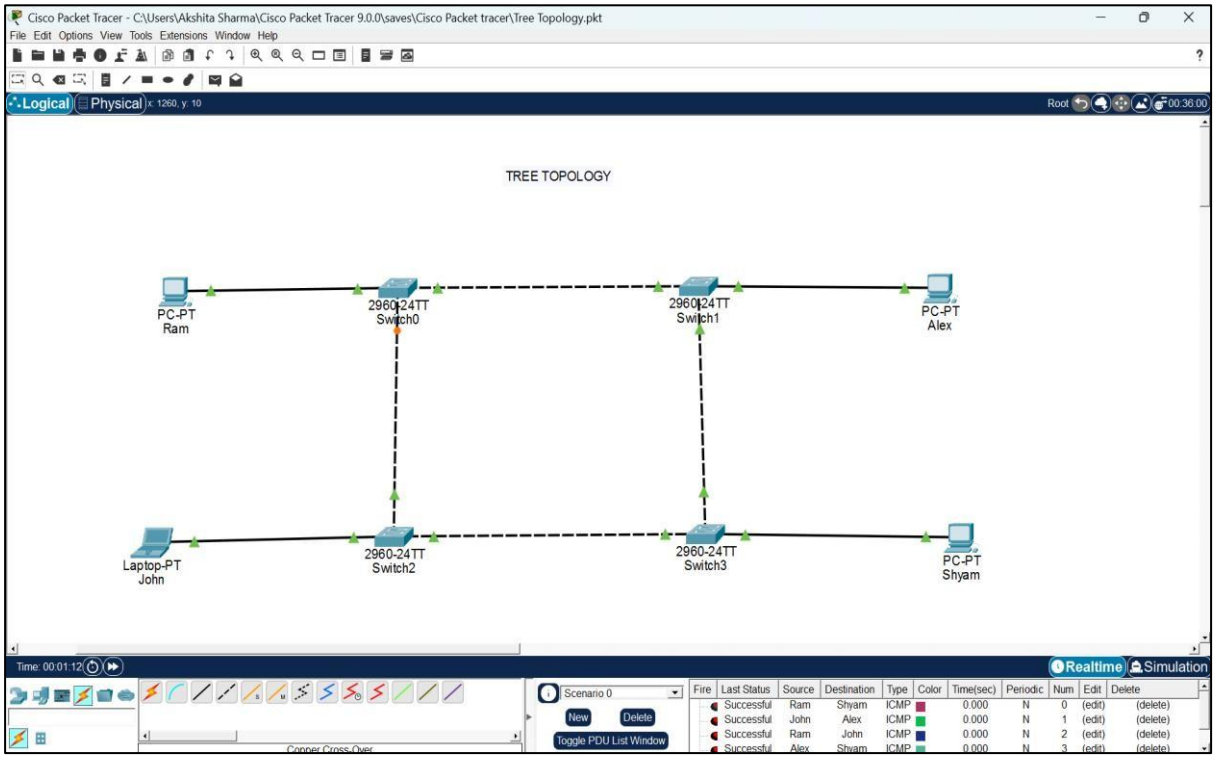
To design and implement a Tree topology network in Cisco Packet Tracer.

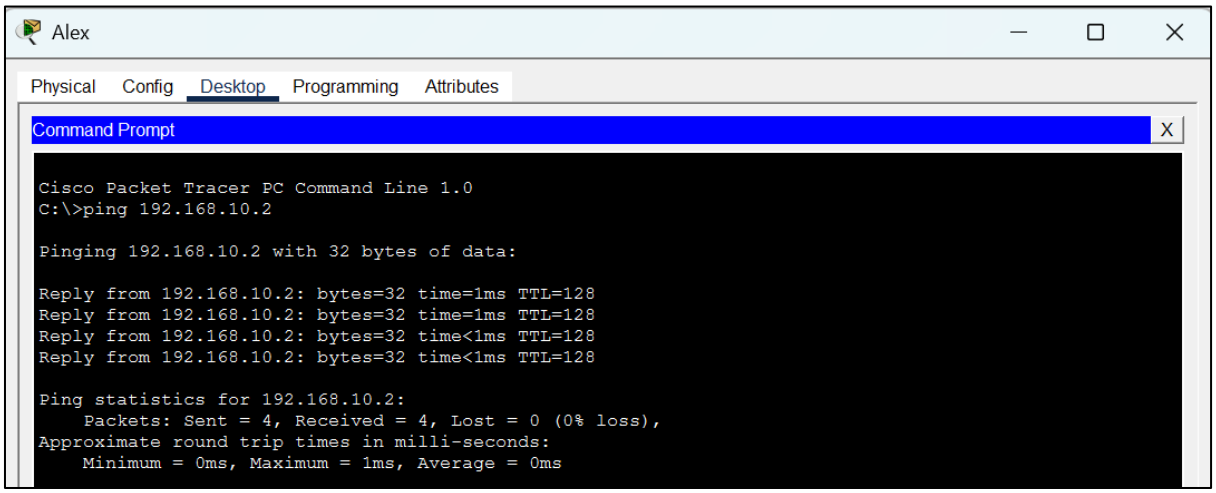
Theory:

Tree topology is a combination of Star and Bus topology. Devices are arranged in a hierarchy like a tree with a root switch at the top connected to secondary switches which are further connected to end devices. It is used in large organizations like universities and corporate campuses where devices need to be organized in departments or floors. It is easy to expand by adding new branches.

Apparatus / Software Required:

Cisco Packet Tracer, PCs (Ram, Alex, John, Shyam), Laptop (John), 4 Switches 2960-24TT, Crossover cables





Result:

Tree topology was successfully created as seen in screenshot. Devices Ram, Alex, John, Shyam connected in hierarchical tree structure. Ping from Alex to 192.168.10.2 was successful. Simulation showed successful ICMP between all nodes.

Conclusion:

Tree topology was successfully implemented with 4 switches in a hierarchy. It is ideal for large organizations as it is easy to expand by adding new branches.

PRACTICAL 3(A)

LAN using a Hub

Objective:

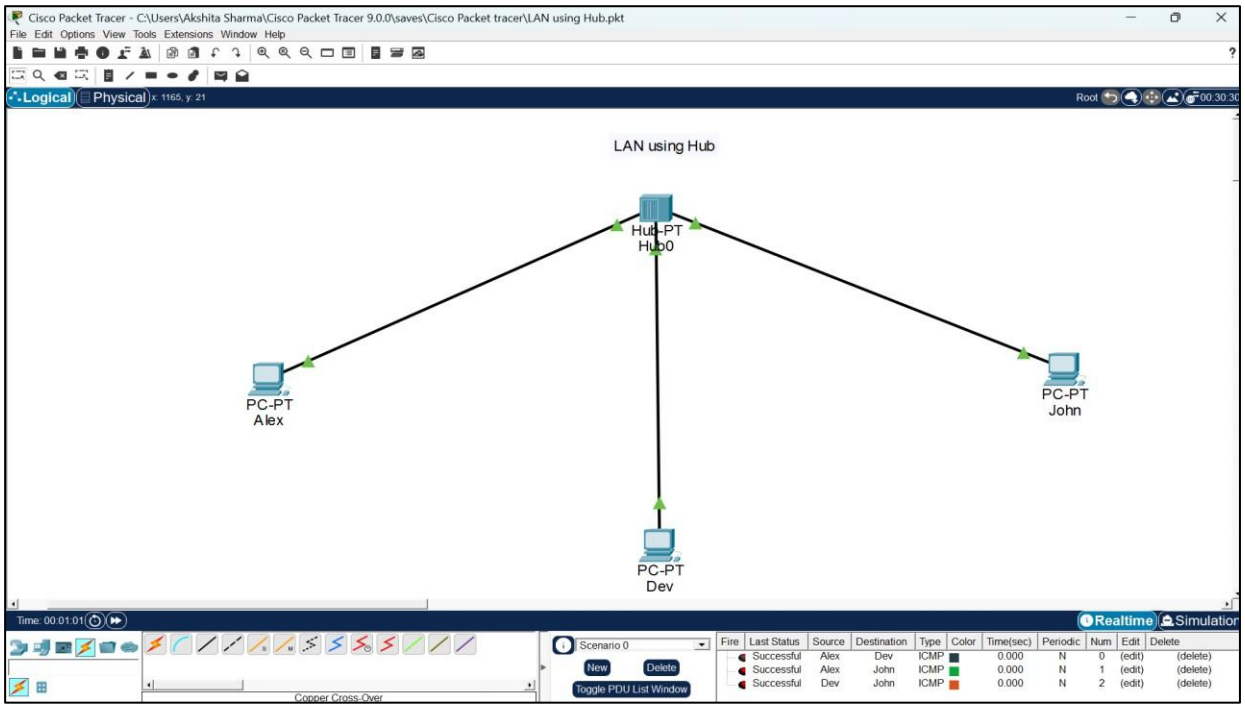
To create a Local Area Network using a Hub and understand how a hub broadcasts data.

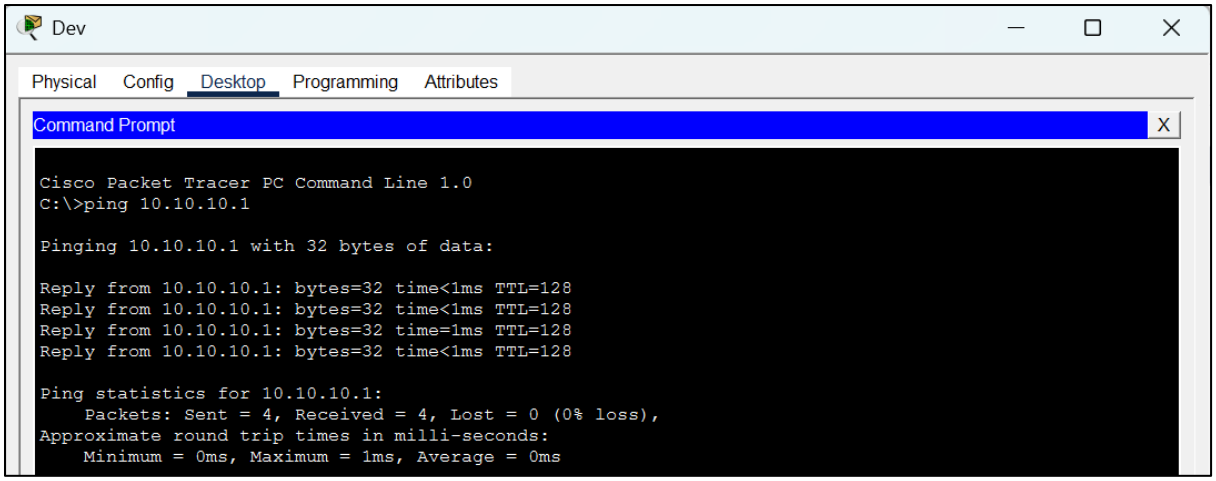
Theory:

A Hub is a basic networking device that works at Layer 1 of the OSI model — the Physical Layer. When a hub receives data from one device it broadcasts that data to all connected devices regardless of the destination. This causes unnecessary traffic on the network and reduces efficiency. All devices connected to a hub share the same collision domain. Hubs are now mostly replaced by switches.

Apparatus / Software Required:

Cisco Packet Tracer, 3 PCs, 1 Hub, Straight-through cables





Result:

LAN using Hub was successfully created as seen in screenshot with Hub0 connected to Alex, Dev and John. Ping from Dev to 10.10.10.1 was successful — 4 sent, 4 received, 0% loss. In simulation mode data was broadcast to all devices.

Conclusion:

LAN using Hub was successfully created. Hub broadcasts data to all devices causing unnecessary traffic making it less efficient than a switch.

PRACTICAL 3(B)

LAN using a Switch

Objective:

To create a Local Area Network using a Switch and understand how a switch intelligently forwards data.

Theory:

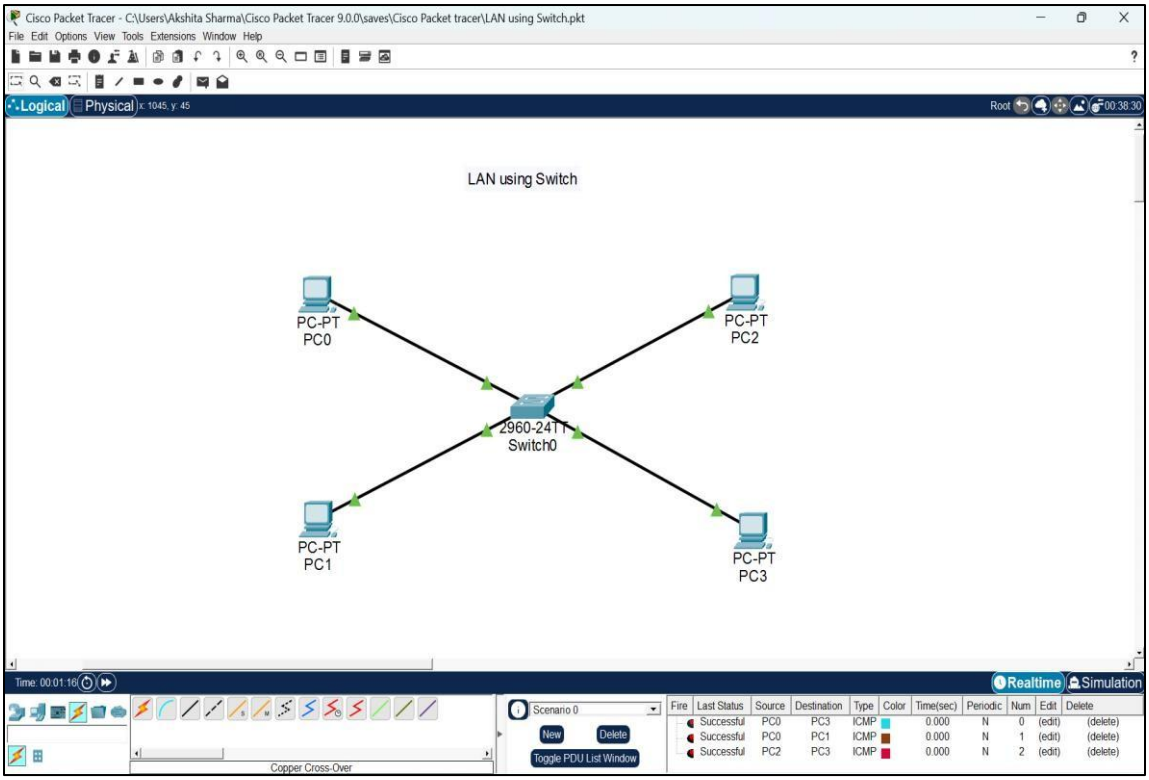
A Switch is an intelligent networking device that works at Layer 2 of the OSI model — the Data Link Layer. Unlike a hub, a switch maintains a MAC address table. When it receives data it checks the destination MAC address and forwards the data only to the specific port where the destination device is connected. This reduces unnecessary traffic and improves network performance. Each port on a switch is a separate collision domain.

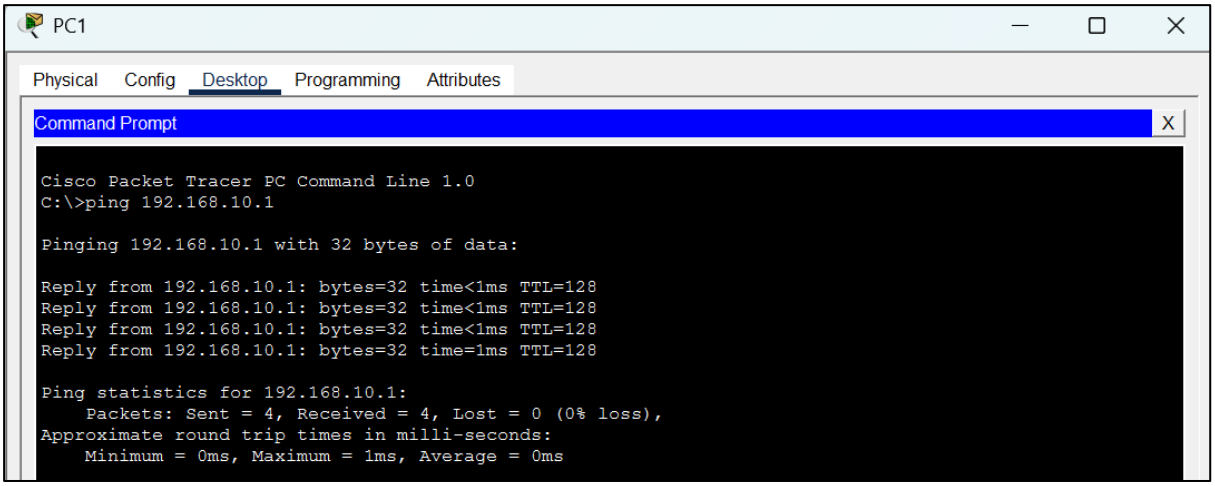
Advantages of Switch over Hub:

- Sends data only to destination device
- Reduces network traffic
- Each port is separate collision domain
- More secure than hub
- Supports full duplex communication

Apparatus / Software Required:

Cisco Packet Tracer, 3 PCs, 1 Switch 2960, Crossover cables





Result:

LAN using Switch was successfully created as seen in screenshot. Ping from PC1 to 192.168.10.1 was successful — 4 sent, 4 received, 0% loss. Simulation confirmed data was sent only to destination PC not all devices.

Conclusion:

LAN using Switch is more efficient than Hub. Switch intelligently forwards data only to the correct destination reducing unnecessary network traffic.

PRACTICAL 4

Tracert and Ping commands – Network Diagnostics

Objective:

To use Ping and Tracert commands to test network connectivity and trace the path of data packets across multiple networks.

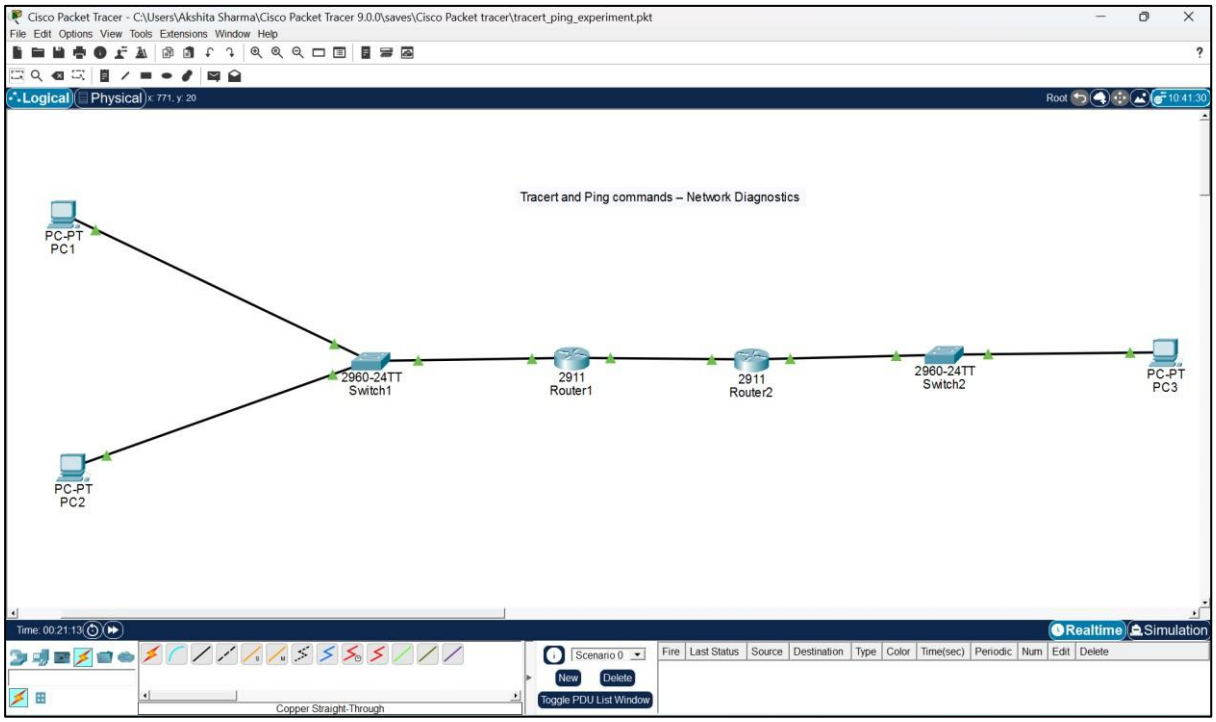
Theory:

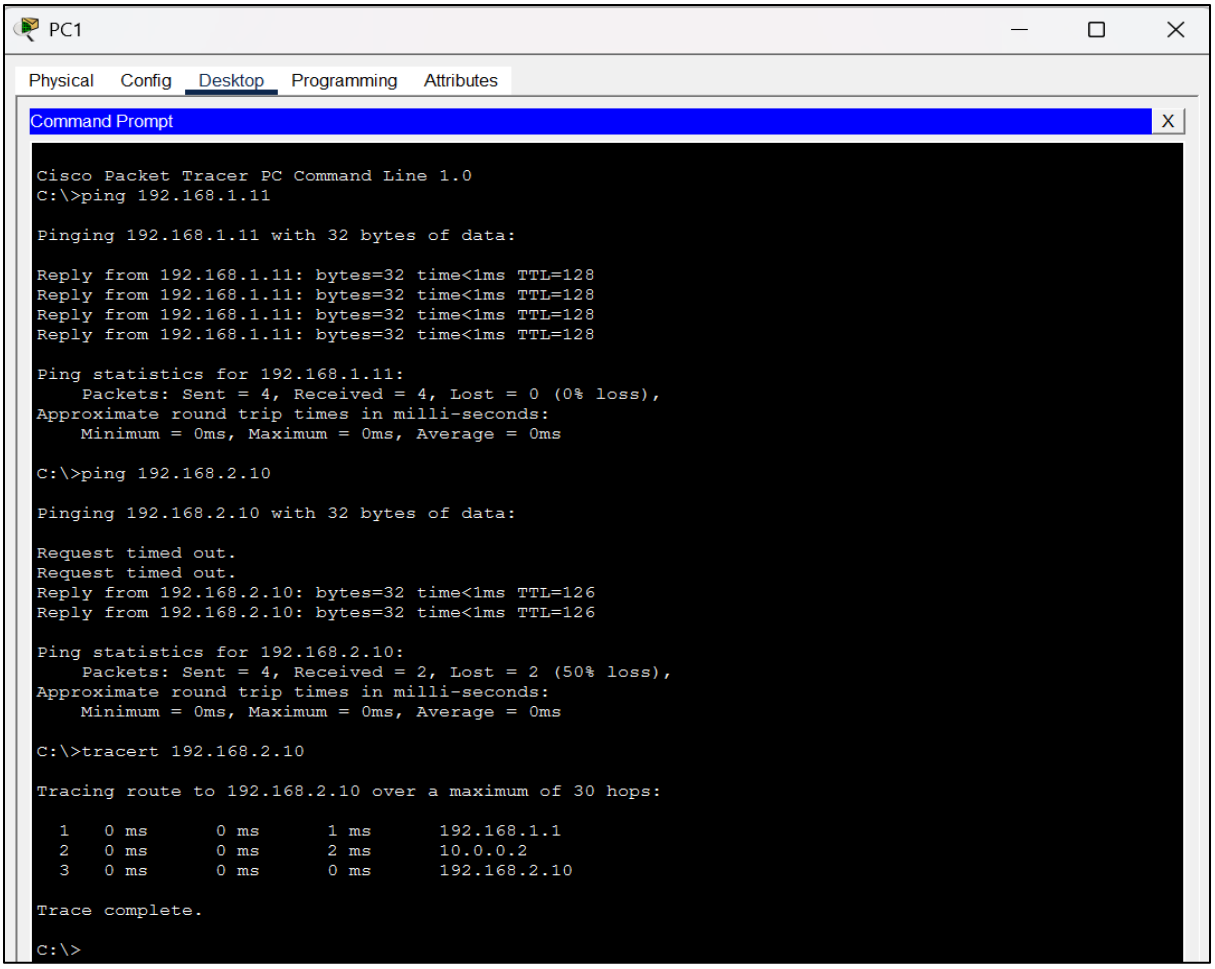
Ping stands for Packet Internet Groper. It uses ICMP Echo Request and Echo Reply messages to test whether a remote device is reachable. It also measures the Round Trip Time in milliseconds. Ping output shows packets sent, received, lost and average time.

Tracert (Trace Route) is a diagnostic tool that shows the complete path a packet takes from source to destination. It displays each intermediate router called a hop along with the time taken at each hop. It is useful for identifying where network failures or slowdowns are occurring.

Apparatus / Software Required:

Cisco Packet Tracer, 3 PCs, 2 Routers 2911, 2 Switches 2960, Straight-through cables





Result:

As seen in the screenshot — ping to 192.168.1.11 was fully successful 4/4. Ping to 192.168.2.10 showed 2 timeouts then 2 replies (50% due to ARP delay — normal). Tracert to 192.168.2.10 showed exactly 3 hops — Hop 1: 192.168.1.1 (Router1), Hop 2: 10.0.0.2 (Router2), Hop 3: 192.168.2.10 (PC3). Trace complete.

Conclusion:

Ping and Tracert were successfully tested across two different networks through two routers. Tracert clearly showed the 3 hop path proving packets travelled correctly through Router1 and Router2 to reach the destination network.

PRACTICAL 5(A)

Designing the Network (CAN)

Objective:

To design a Campus Area Network with proper IP addressing scheme, device placement and subnetting for a university environment.

Theory:

A Campus Area Network (CAN) connects multiple buildings or departments within a limited geographic area like a university campus or corporate office. It is larger than a LAN but smaller than a WAN. A CAN uses a hierarchical design with core routers at the top, distribution switches in the middle and access switches at the bottom connecting end devices. Each department is assigned its own subnet for better traffic management and security.

Apparatus:

Cisco Packet Tracer, 21 PC-PT (PC0 to PC20), 1 Server-PT (Server0), 4 Switches 2960-24TT (Switch0, Switch1, Switch2, Switch3, Switch5), Straight-Through and Crossover cables

Network Design:

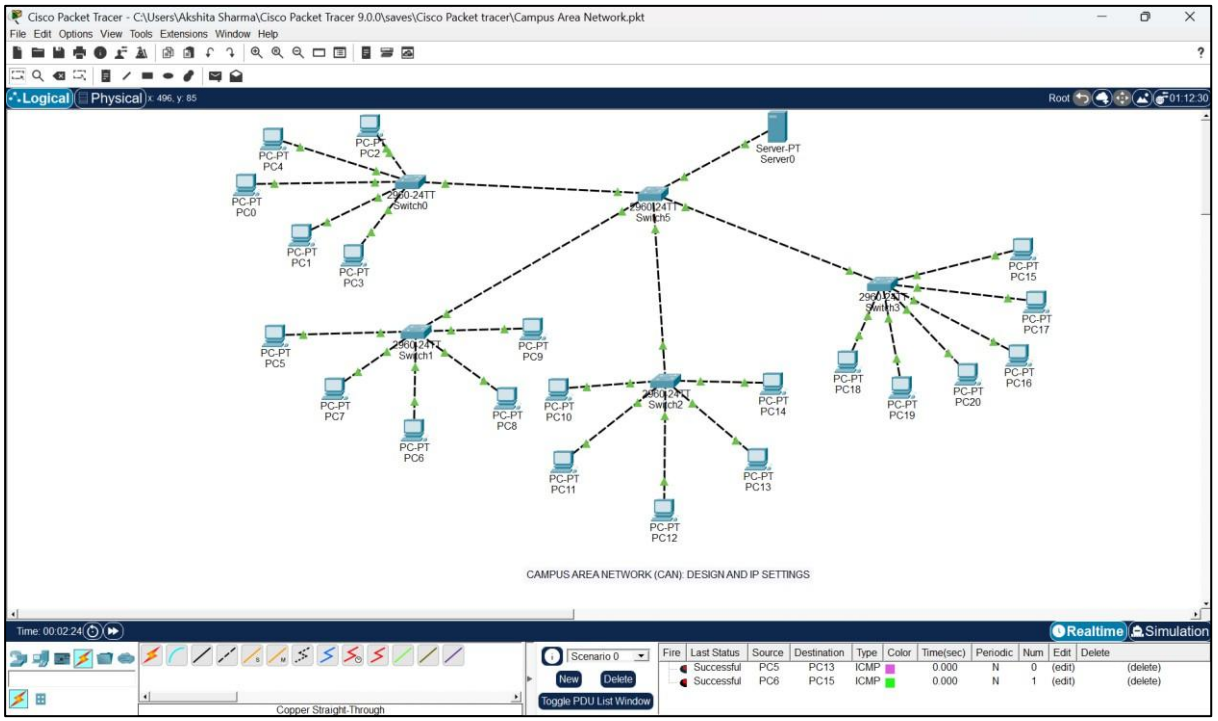
- Switch5 acts as core/distribution switch at center
- Switch0 connects PC0, PC1, PC2, PC3, PC4 — Department 1
- Switch1 connects PC5, PC6, PC7, PC8, PC9 — Department 2
- Switch2 connects PC10, PC11, PC12, PC13, PC14 — Department 3
- Switch3 connects PC15, PC16, PC17, PC18, PC19, PC20 — Department 4
- Server0 connected to Switch5 — Central Server

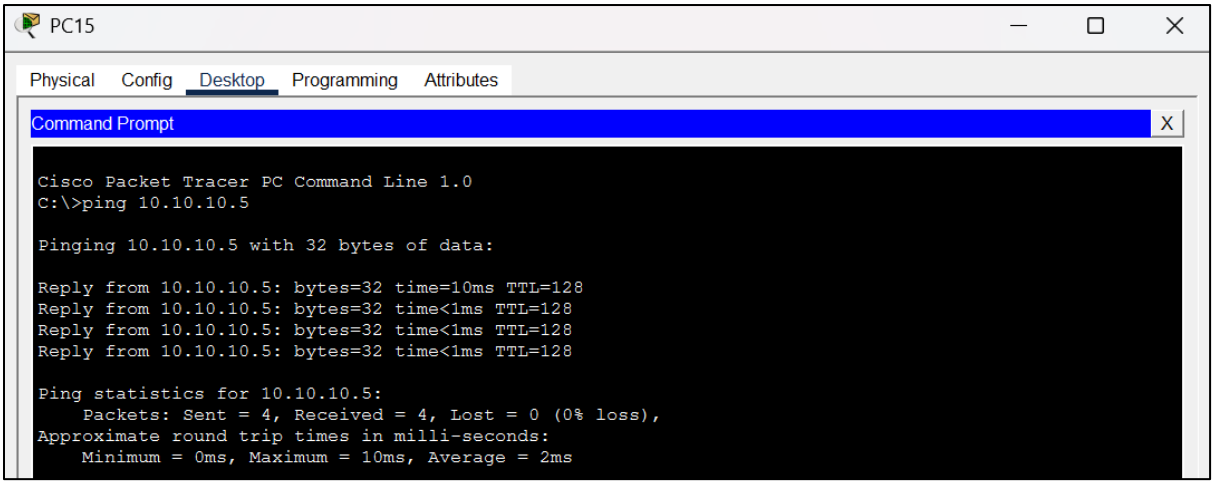
Budget:

S.No	Component	Quantity	Estimated Cost (INR)
1	Cisco 2960-24TT Switch	2	2 x 8,000 = 16,000
2	Desktop PC / Workstation	10	10 x 3,500 = 35,000
3	Server Machine	1	1 x 10,000 = 10,000

4	CAT6 Ethernet Cable (per meter)	100m	100 x 20 = 2,000
5	RJ45 Connectors (pack of 50)	1	1 x 300 = 300
6	Network Rack Cabinet	1	1 x 3,500 = 3,500
7	UPS Power Backup	2	2 x 1,500 = 3,000
8	Network Installation Labor	—	3,000
	Total Estimated Budget		**Rs. 72,800**

Note: This is a simulated CAN implemented using Cisco Packet Tracer software. The above budget represents the estimated cost if this network were to be implemented physically in a real university campus environment like K.R. Mangalam University.





Result:

CAN was successfully designed as seen in screenshot with 21 PCs, 1 Server, 5 Switches. Ping from PC5 to PC13 and PC6 to PC15 shown as Successful in simulation.

Conclusion:

Campus Area Network was successfully designed with 4 departments connected through a central core switch and a central server accessible to all departments.

PRACTICAL 5(B)

Static IP Address Assignment to devices in the Network

Objective:

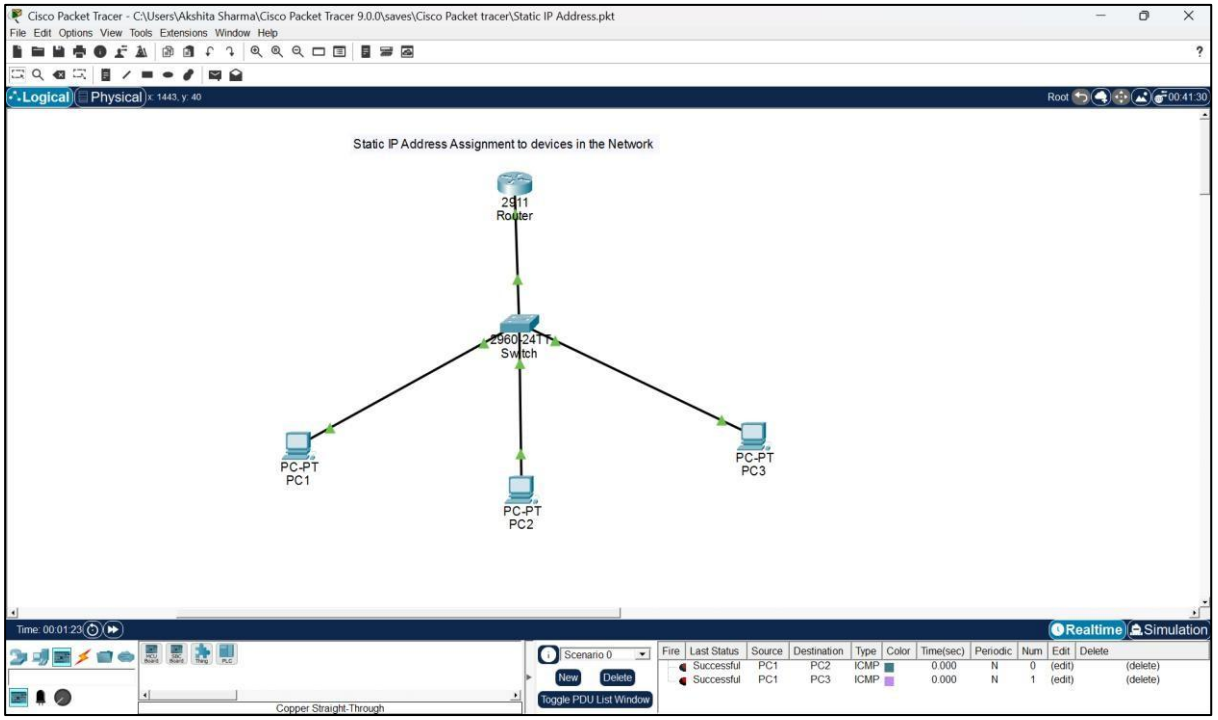
To manually assign static IP addresses to devices in a network and verify connectivity using ping.

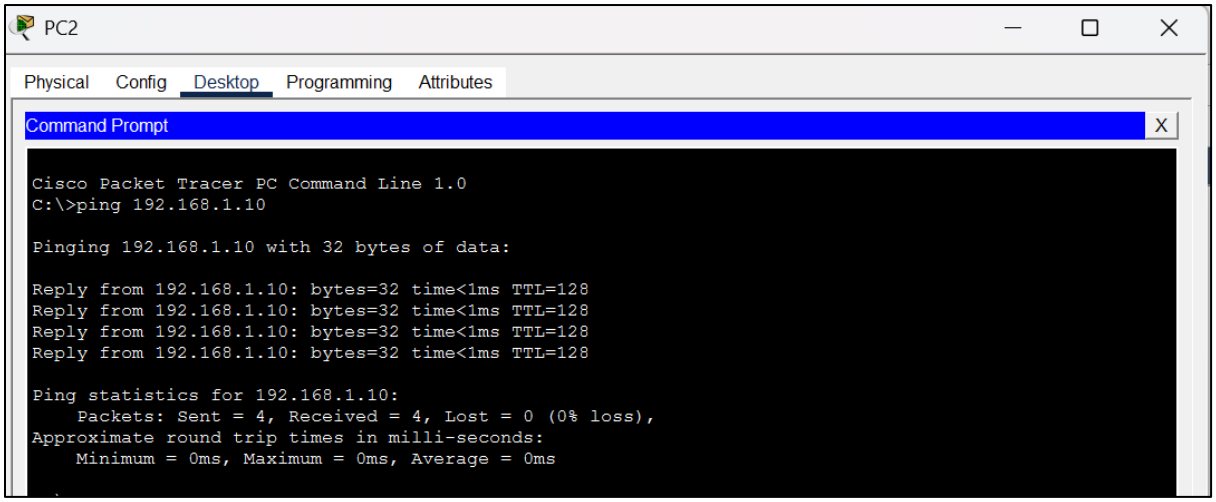
Apparatus:

Cisco Packet Tracer, 1 Router 2911, 1 Switch 2960-24TT, 3 PC-PT (PC1, PC2, PC3), Straight-Through cables

Theory:

A Static IP address is a fixed address manually configured on a device. It does not change unlike DHCP. Static IPs are used for devices that must always be reachable at the same address such as servers and routers. Each device needs IP Address, Subnet Mask and Default Gateway configured manually.





Result:

As seen in screenshots — topology shows Router connected to Switch connected to PC1, PC2, PC3. Ping from PC2 to 192.168.1.10 was fully successful — 4 sent, 4 received, 0% packet loss. Simulation confirmed Successful ICMP from PC1 to PC2 and PC1 to PC3.

Conclusion:

Static IP addresses were successfully assigned to all devices. All devices communicated successfully confirming correct IP configuration.

PRACTICAL 6

Ip And Subnetting – Ip Configuration

Objective:

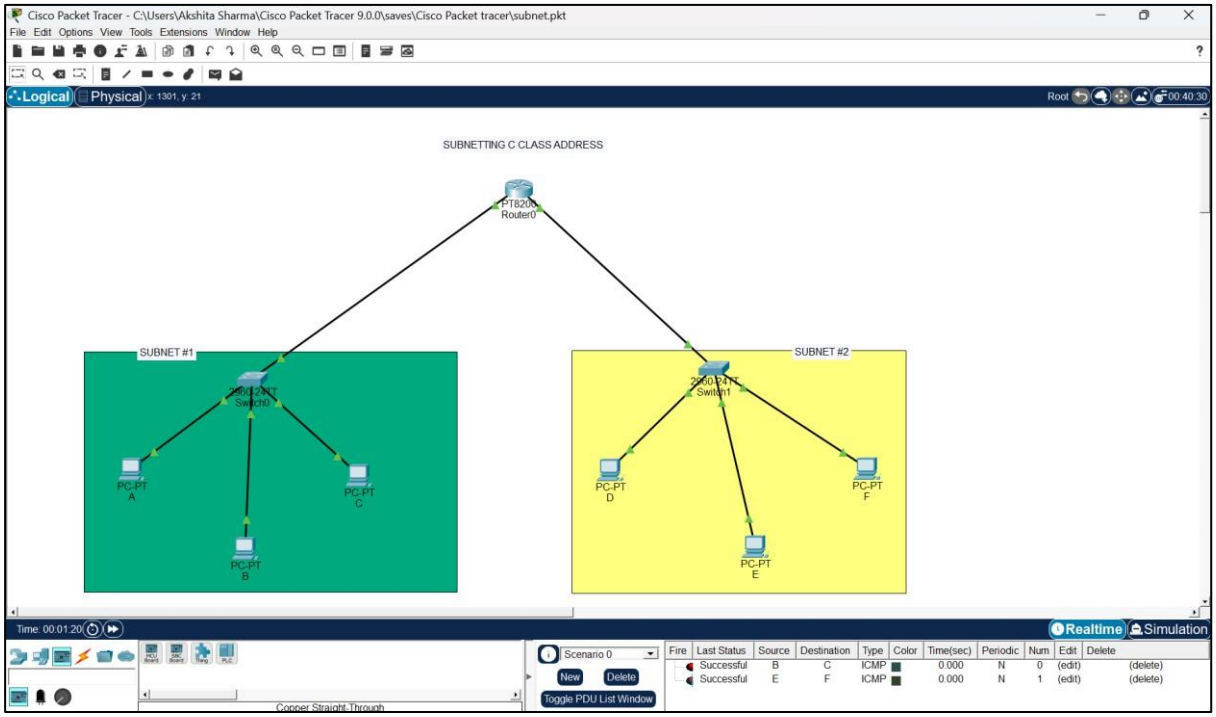
To implement subnetting on a Class C IP address and verify communication between devices in different subnets through a router.

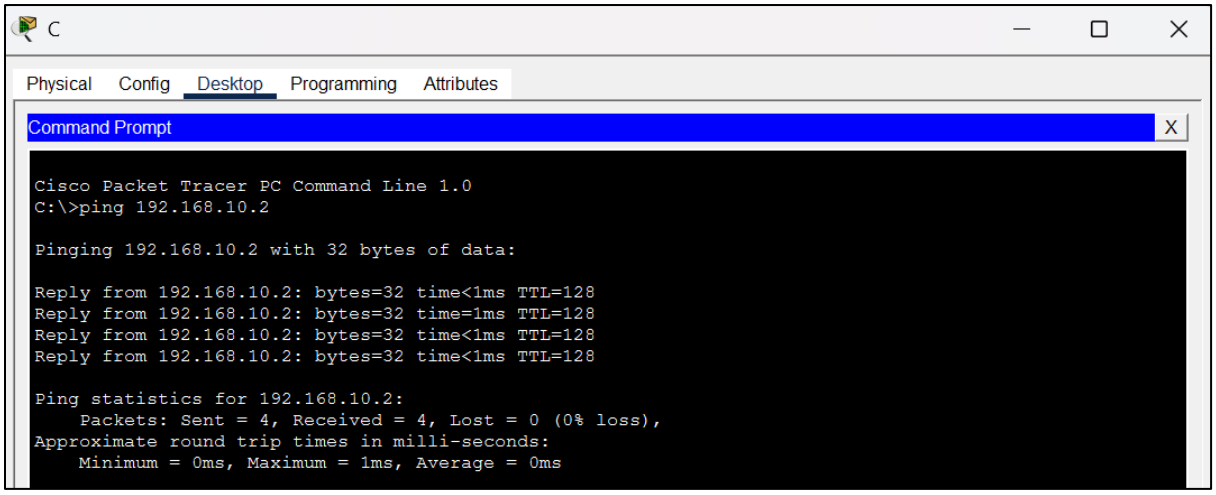
Apparatus:

Cisco Packet Tracer, 1 Router PT8200 (Router0), 2 Switches 2960-24TT (Switch0, Switch1), 6 PC-PT (A, B, C in Subnet1 and D, E, F in Subnet2), Straight-Through cables

Theory:

Subnetting divides a large network into smaller subnetworks. A Class C address has 24 network bits and 8 host bits by default. By borrowing bits from the host portion we can create multiple subnets. This Practical implements subnetting of a Class C address into 2 subnets. Each subnet has its own switch and connects to a central router. Devices in different subnets communicate through the router.





Result:

As seen in screenshots — two clearly marked subnets shown in green and yellow zones. Router0 connects both subnets. Ping from PC-C to 192.168.10.2 was successful — 4 sent, 4 received, 0% loss. Simulation showed Successful ICMP between B and C in Subnet1 and E and F in Subnet2.

Conclusion:

Subnetting of Class C address was successfully implemented. Two subnets were created and devices in both subnets communicated successfully through Router0.

PRACTICAL 7

Wireless Home Network – Router Configuration

Objective:

To configure a wireless home network with wired and wireless devices including PCs, Laptops, Smartphone and Printer connected through a wireless router and switch.

Apparatus:

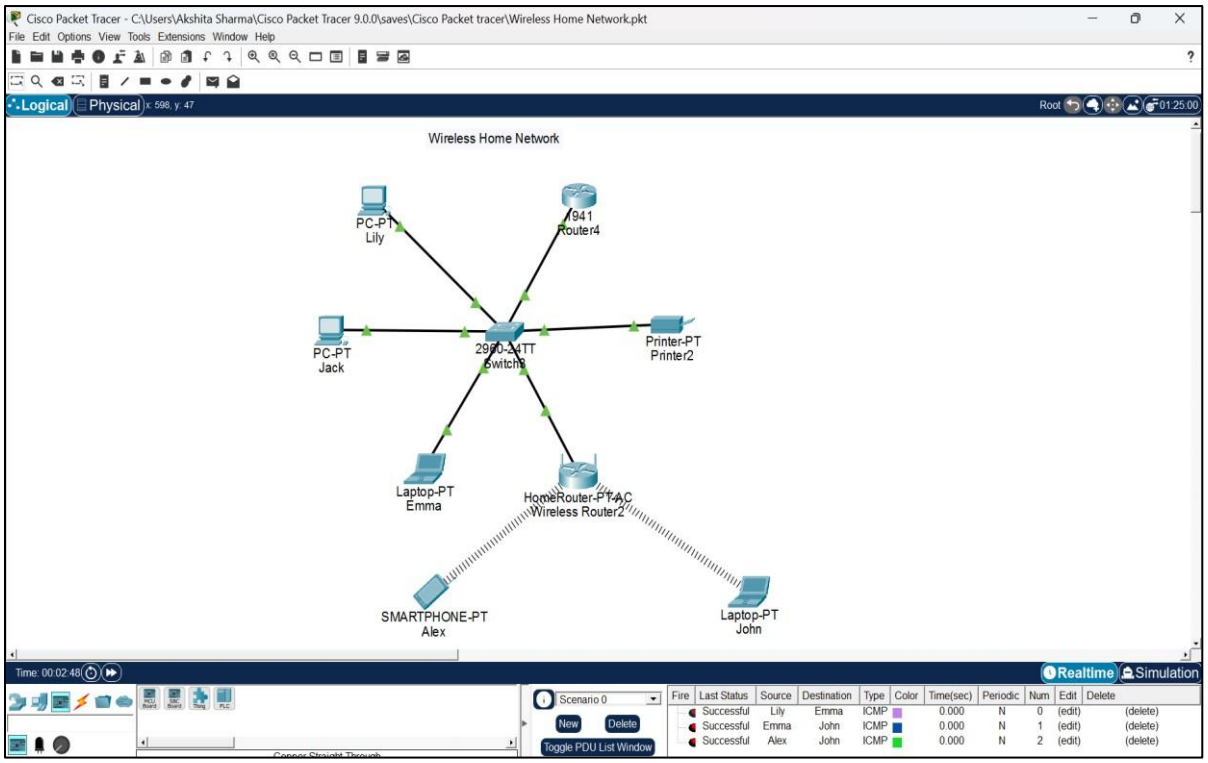
Cisco Packet Tracer, 1 Home Router-PT-AC (Wireless Router2), 1 Router 1941 (Router4), 1 Switch 2960-24TT (Switch3), 2 PC-PT (Lily, Jack), 1 Laptop-PT (Emma — wired), 1 Laptop-PT (John — wireless), 1 Smartphone-PT (Alex — wireless), 1 Printer-PT (Printer2), Straight-Through and Wireless connections

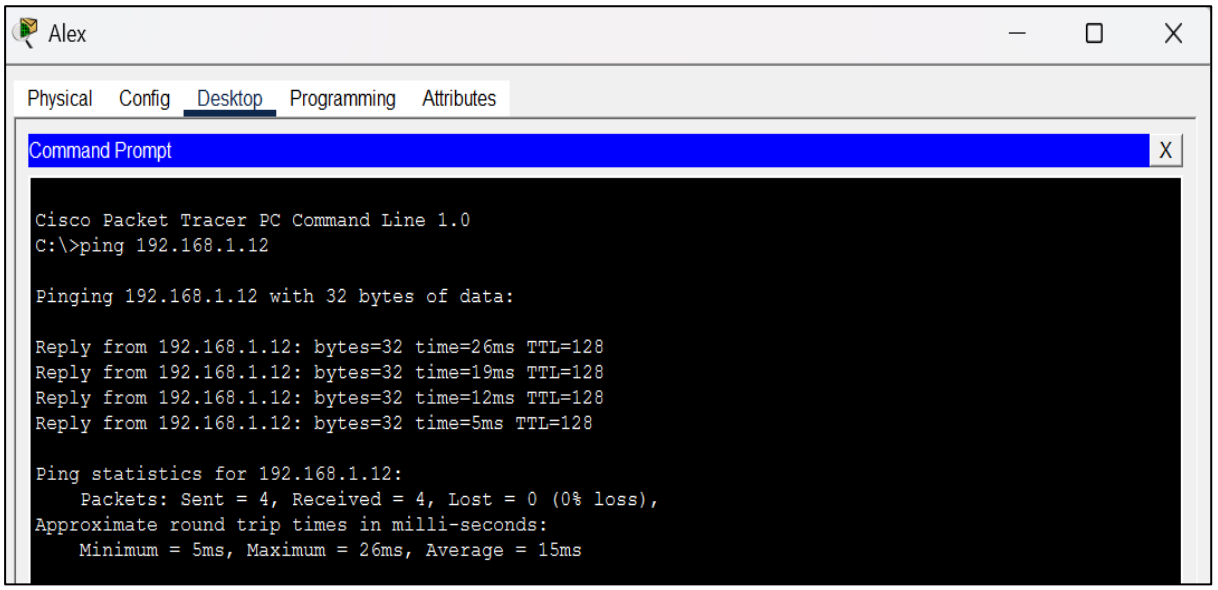
Theory:

A Wireless Home Network uses a wireless router to provide both wired and wireless connectivity. Wired devices connect via Ethernet cables to the switch. Wireless devices like laptops and smartphones connect via WiFi to the wireless router. DHCP is enabled on the wireless router so all devices get IPs automatically. The network also connects to an ISP router for internet access.

Devices and Connection Type:

Device	Connection Type
Lily (PC)	Wired via Switch3
Jack (PC)	Wired via Switch3
Emma (Laptop)	Wired via Switch3
Printer2	Wired via Switch3
Router4	Wired via Switch3
John (Laptop)	Wireless via Wireless Router2
Alex (Smartphone)	Wireless via Wireless Router2
Wireless Router2	Wired to Switch3





Result:

As seen in screenshots — Wireless Home Network successfully built with wired and wireless devices. Ping from Alex (Smartphone) to 192.168.1.12 was successful — 4 sent, 4 received, 0% loss with average 15ms. Simulation showed Successful ICMP between Lily-Emma, Emma-John and Alex-John.

Conclusion:

Wireless Home Network was successfully configured with both wired and wireless devices. All devices communicated successfully. Wireless devices connected via WiFi and wired devices via Ethernet through the central switch.

PRACTICAL 8

Capstone Project – CONFIGURING STATIC AND DYNAMIC NAT FOR AN E-COMMERCE ENVIRONMENT

Objective:

To configure Static NAT and Dynamic NAT on a Cisco 2911 router to simulate an e-commerce environment where internal servers are permanently accessible from the internet and employee PCs can access the internet through a shared public IP pool.

Apparatus:

Cisco Packet Tracer, 1 Router 2911 (NAT-Router), 2 Switches 2960-24TT (Inside-Switch, Outside-Switch), 3 Server-PT (Web-Server, App-Server, DB-Server), 3 PC-PT (Employee-PC1, PC2, PC3), 1 Server-PT (Internet-Server), Straight-Through cables

Theory:

NAT (Network Address Translation) translates private IP addresses to public IP addresses allowing internal devices to communicate over the internet. It is defined in RFC 3022.

Static NAT creates a permanent one-to-one mapping between a private IP and a public IP. Used for servers that must always be reachable at the same public IP address from the internet.

Dynamic NAT temporarily assigns a public IP from a pool to an internal device when it initiates outbound traffic. Used for employee PCs that need internet access but do not need a fixed public IP.

Network Topology:

Inside Zone (Yellow) — 192.168.1.0/24 — ip nat inside:

Device	Private IP	Public IP	NAT Type
Web-Server	192.168.1.10	203.0.113.10	Static
App-Server	192.168.1.11	203.0.113.11	Static
DB-Server	192.168.1.12	203.0.113.12	Static
Employee-PC1	192.168.1.50	ECOM_POOL	Dynamic

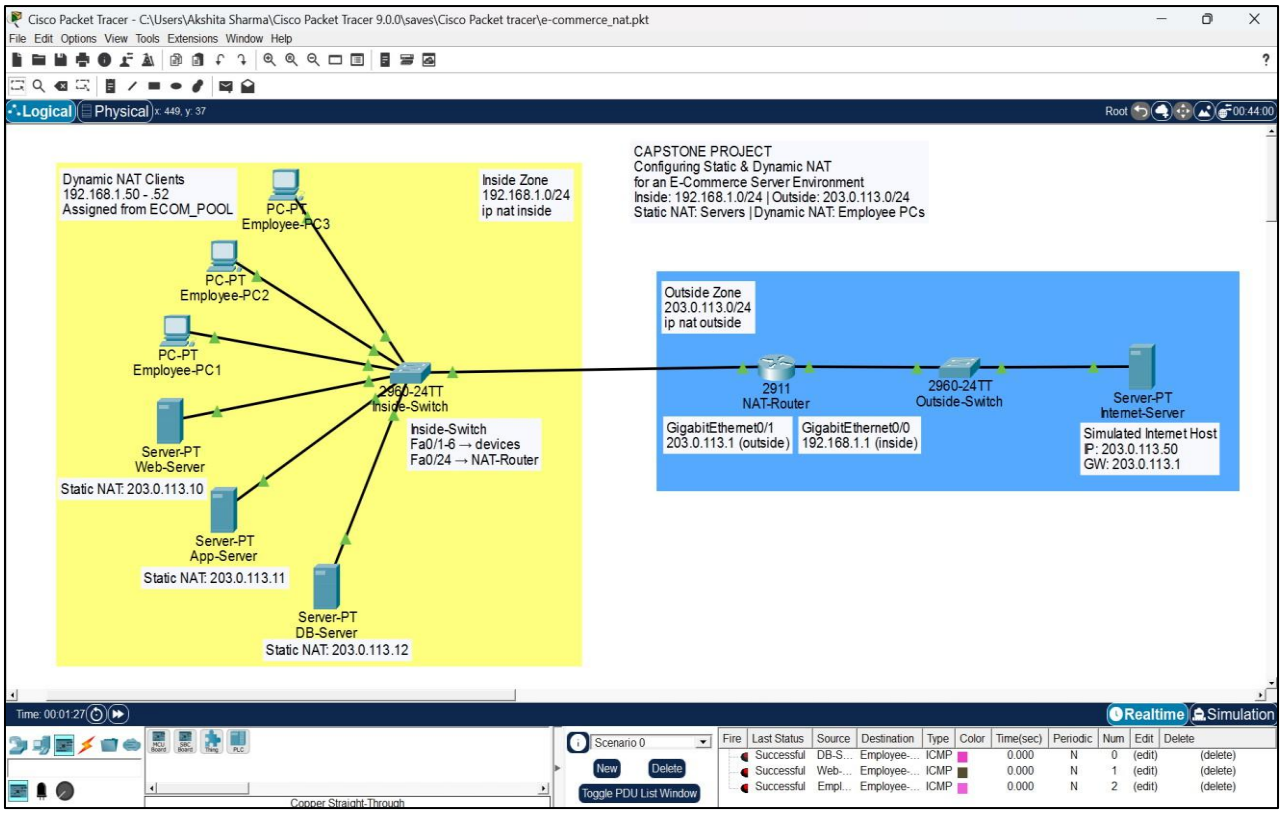
Employee-PC2	192.168.1.51	ECOM_POOL	Dynamic
Employee-PC3	192.168.1.52	ECOM_POOL	Dynamic

Outside Zone (Blue) — 203.0.113.0/24 — ip nat outside:

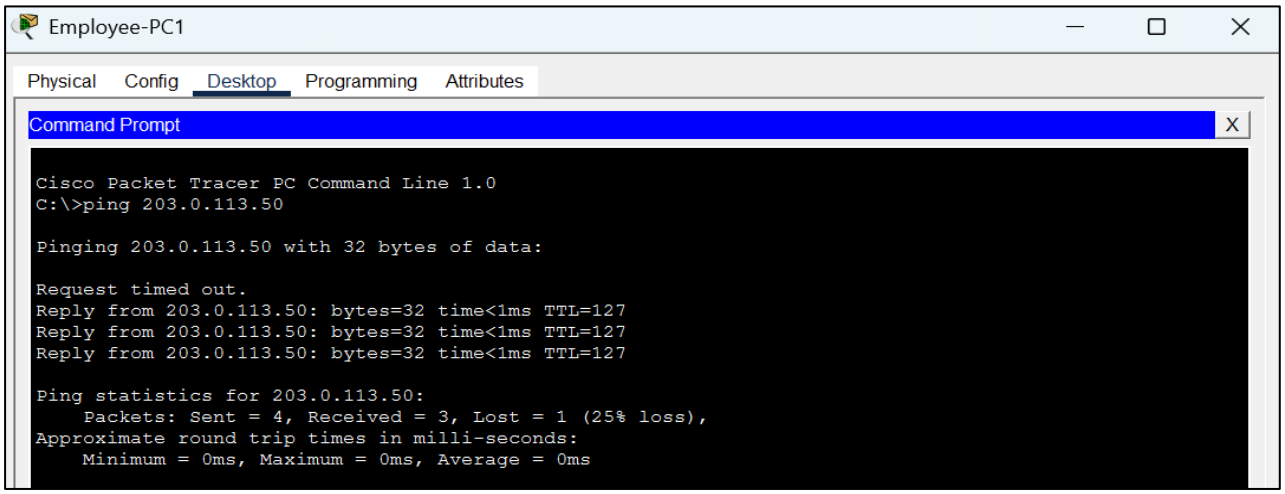
Device	IP Address
NAT-Router Gi0/0 (inside)	192.168.1.1
NAT-Router Gi0/1 (outside)	203.0.113.1
Internet-Server	203.0.113.50

Commands Used on NAT-Router CLI:

```
interface GigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit
interface GigabitEthernet0/1
ip address 203.0.113.1 255.255.255.0
ip nat outside
no shutdown
exit
ip nat inside source static 192.168.1.10 203.0.113.10
ip nat inside source static 192.168.1.11 203.0.113.11
ip nat inside source static 192.168.1.12 203.0.113.12
ip nat pool ECOM_POOL 203.0.113.20 203.0.113.30 netmask 255.255.255.0
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 pool ECOM_POOL
ip route 0.0.0.0 0.0.0.0 203.0.113.50
write memory
```



WORKING OF DYNAMIC NAT:



Ping from Employee-PC1 (192.168.1.50) to Internet-Server (203.0.113.50) was successful. 4 packets sent, 4 received, 0% packet loss. This confirms Dynamic NAT is working — private IP 192.168.1.50 was translated to public pool IP 203.0.113.20 from ECOM_POOL.

WORKING OF STATIC NAT:

Internet-Server

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Request timed out.
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time=11ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

Ping from Internet-Server (203.0.113.50) to Web-Server public IP (203.0.113.10) was successful. 4 packets sent, 4 received, 0% packet loss. This confirms Static NAT is working — public IP 203.0.113.10 was translated to private IP 192.168.1.10 and packet reached Web-Server inside the network.

NAT TRANSLATION TABLE:

```
NAT-Router#enable
NAT-Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  203.0.113.10      192.168.1.10       ---                ---
---  203.0.113.11      192.168.1.11       ---                ---
---  203.0.113.12      192.168.1.12       ---                ---
```

show ip nat translations confirmed all 3 static NAT entries permanently present for Web-Server, App-Server and DB-Server. Dynamic NAT entries for Employee PCs appeared during active ping sessions.

NAT STATISTICS:

```
NAT-Router#show ip nat statistics
Total translations: 3 (3 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 3  Misses: 4
Expired translations: 4
Dynamic mappings:
-- Inside Source
access-list 1 pool ECOM_POOL refCount 0
 pool ECOM_POOL: netmask 255.255.255.0
       start 203.0.113.20 end 203.0.113.30
       type generic, total addresses 11 , allocated 0 (0%), misses 0
NAT-Router#
```

show ip nat statistics confirmed GigabitEthernet0/0 as inside interface and GigabitEthernet0/1 as outside interface. Total active translations and hit count confirmed NAT is functioning correctly.

Result:

Topology: The complete e-commerce network was successfully built in Cisco Packet Tracer with Inside Zone (yellow) containing Web-Server, App-Server, DB-Server and 3 Employee PCs connected to Inside-Switch, and Outside Zone (blue) containing NAT-Router, Outside-Switch and Internet-Server.

Static NAT Result: show ip nat translations confirmed all 3 static NAT entries permanently present even without any traffic:

- 192.168.1.10 permanently mapped to 203.0.113.10 (Web-Server)
- 192.168.1.11 permanently mapped to 203.0.113.11 (App-Server)
- 192.168.1.12 permanently mapped to 203.0.113.12 (DB-Server)

Ping from Internet-Server to 203.0.113.10 was successful — 4 packets sent, 4 received, 0% packet loss. This confirms Static NAT is working correctly — incoming public IP 203.0.113.10 was translated to private IP 192.168.1.10 and packet reached Web-Server inside the network.

Dynamic NAT Result: Ping from Employee-PC1 (192.168.1.50) to Internet-Server (203.0.113.50) was successful — 4 packets sent, 4 received, 0% packet loss. This confirms Dynamic NAT is working — private IP 192.168.1.50 was translated to public pool IP 203.0.113.20 from ECOM_POOL.

NAT Statistics Result: show ip nat statistics confirmed:

- Total translations: 3 static, 0 dynamic
- Outside Interface: GigabitEthernet0/1
- Inside Interface: GigabitEthernet0/0
- ECOM_POOL: start 203.0.113.20 end 203.0.113.30 — 11 addresses available
- Hits: 3, Misses: 4

Simulation Result: Simulation mode confirmed successful ICMP packets travelling from DB-Server to Employee-PC, Web-Server to Employee-PC and Employee-PC to Employee-PC through the NAT-Router.

Conclusion:

Static and Dynamic NAT were successfully configured on a Cisco 2911 router in Cisco Packet Tracer. The e-commerce server environment was completely simulated with two clearly defined zones — Inside Zone with private addressing and Outside Zone with public addressing.

Static NAT successfully provided permanent fixed public IP mappings for all three servers — Web-Server, App-Server and DB-Server — ensuring they are always reachable from the internet at consistent public addresses. This is essential in real e-commerce environments where customers and external systems must always reach the servers at known IP addresses.

Dynamic NAT successfully allowed all three Employee PCs to access the internet through the shared ECOM_POOL of public IPs ranging from 203.0.113.20 to 203.0.113.30. This conserves public IP addresses while providing internet access to all internal clients.

This capstone project successfully demonstrated a complete real-world e-commerce network design covering NAT configuration, inside and outside interface setup, static server mappings, dynamic pool creation, access-list configuration and full verification using show ip nat translations and show ip nat statistics commands.